Substrate Carries, Architecture Pays: Check-In-Level Embeddings for Multi-Task POI Prediction

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Abstract. Multi-task learning for point-of-interest prediction typically pairs joint next-category and next-region heads over a shared graph embedding, assuming joint training transfers signal across tasks. We introduce Check2HGI, a check-in-level contextual graph embedding extending HGI's hierarchical contrastive objective with per-visit emission rate, and run a controlled five-state Gowalla ablation under user-disjoint folds (transductive substrate). The substrate is task-asymmetric: per-visit context lifts single-task next-category macro-F1 by 14 to 29 percentage points at every state (paired Wilcoxon p of 0.0312), while per-place embeddings tie or marginally exceed it under matched-head ceilings on next-region. Joint multi-task training over a cross-attention backbone then delivers a category lift at four of five states; the cost is a sign-consistent 7 to 17 percentage point drop on next-region top-10 accuracy, the textbook tradeoff. A per-visit counterfactual at Alabama and Arizona attributes about 72 and 64 percent of the substrate gain to per-visit context.

Keywords: Multi-Task Learning · Point-of-Interest Prediction · Graph Embeddings.

1 Introduction

Location-Based Social Networks (LBSNs) record user check-ins at scale [4], motivating two recurring Point-of-Interest (POI) prediction tasks: the next visit's semantic category, which feeds content-aware recommendation and intent inference, and its spatial region, which feeds demand forecasting and short-horizon mobility planning. The two tasks place disjoint demands. **Category prediction** is a coarse semantic decision shaped by per-visit context (intent, time-of-day, co-location). **Region prediction** is a fine-grained spatial ranking that hinges on spatial-temporal regularities and short-horizon transitions. The multi-task learning (MTL) appeal is standard: a shared representation should let one head's signal transfer and lift both [2,13].

Two recent works frame the bottleneck. The first reported hard-parameter-sharing MTL on POI-stable graph embeddings (DGI [14]) yielding only marginal gains within standard deviations; the diagnosis was representation mismatch [12].

A follow-up decomposed the monolithic embedding into spatial, temporal, and categorical sub-encoders, recovering category-side gains across three states [10]. Together these suggest embedding choice, not the MTL recipe, is the central factor.

Both prior responses treat the embedding as a *what* axis. We test an orthogonal *granularity* axis: emission rate. POI-stable embeddings like HGI [5] emit one vector per place; a check-in-level variant emits one per visit, capturing intent shifts across two visits to the same café at different hours. Our hypothesis is that **per-visit context** is the load-bearing substrate property for the easier head, and that once the substrate is fixed, joint training behaves like the textbook MTL tradeoff. The carry is task-asymmetric: per-visit *carries* next-category; per-place embeddings tie or marginally exceed it on next-region.

We introduce Check2HGI, a check-in-level extension of the hierarchical graph-infomax line, and run a controlled ablation on five Gowalla U.S.-state splits, Alabama (AL), Arizona (AZ), Florida (FL), California (CA) and Texas (TX), under user-disjoint folds (transductive substrate; §4.1). We compare Check2HGI against HGI under matched-head single-task (STL) ceilings, and run a cross-attention multi-task backbone whose per-task heads match the ceiling classes so that any difference between multi-task and single-task rows is attributable to the sharing architecture rather than head capacity. A per-visit counterfactual decomposes the substrate gain itself; the architectural details, the matched-head construction, and an encoder-frozen isolation protocol that addresses a cross-attention pitfall are deferred to Sections 3–6.

The main contributions of this paper are as follows.

- C0. *Check2HGI*, a check-in-level contextual graph embedding extending HGI's hierarchical contrastive objective with a fourth emission level (Check-in → POI → Region → City), where the same place visited twice can yield two distinct vectors.
- C1. A controlled five-state matched-head ablation that separates substrate from head capacity. Check2HGI lifts single-task category macro-F1 over per-place HGI by +14 to +29 percentage points (pp) at every state (paired Wilcoxon $p = 0.0312$ each, 5/5 folds positive); on next-region the same substrate ties or marginally trails per-place embeddings. A pooled-vs-canonical counterfactual replicates at Alabama and Arizona, attributing ∼72 % and ∼64 % of the category gap to per-visit context.
- C2. A direct characterisation of the multi-task tradeoff under the substrate established by C1, including drop-in balancers and a head-side fix that do not close the next-region gap. This indicates a structural cost of the cross-attention shared-backbone interaction with the region head, rather than a loss-recipe artefact. With the substrate fixed, joint cross-attention multi-task training adds a small category lift at four of five states and pays a sign-consistent 7 to 17 pp cost on next-region Acc@10 at every state.

To support reproducibility, we release the full artefact set (code, fold splits, seeds, per-fold transition priors, and regression tests) at `https://anonymous.4open.science/r/PoiMtlNet-FE6A/`.

Sections 2, 3, and 4 cover related work, method, and setup. Section 5 shows that per-visit context is task-asymmetric: it carries next-category at every state, while per-place embeddings tie or marginally trail it on next-region. With the substrate fixed, joint multi-task training delivers a small category lift at four of five states and pays the textbook region cost. Section 6 attributes the substrate gain through a per-visit counterfactual, ablates drop-in balancers and a head-side fix, and adds a methodological note on cross-attention loss-side ablations. Sections 7 and 8 close.

2 Related Work

Three lines of work bear on our study: graph-based POI embeddings that supply the substrate, multi-task learning balancers that sit on top of any joint architecture, and prior multi-task work on POI prediction tasks. We summarise each line briefly and position our contribution against it; a fourth note on cross-attention multi-task models is deferred to the methodological discussion in Section 6.

2.1 Graph-based POI embeddings

Substrate choice is the load-bearing axis our study tests. Early POI representations learn one vector per place from co-visit statistics, treating the POI as the atomic unit, and category-aware extensions inject category supervision into the same objective. Self-supervised graph contrastive methods then sharpened the representation: Deep Graph Infomax (DGI) [14] maximises mutual information between local node embeddings and a global graph summary, and Hierarchical Graph Infomax (HGI) [5] adapts the contrast across POI, region, and city graphs to capture multi-scale urban structure. DGI and HGI both emit *POI-stable* vectors: the same place visited twice yields the same representation. The substrate we use, *Check2HGI*, is a check-in-level extension of HGI's contrastive objective: contrastive views are sampled at the granularity of a single check-in, and the visit's temporal and co-visit context perturbs the view, so the same POI visited twice can yield two distinct vectors. This per-visit emission rate is the property our substrate axis tests; the recipe is described in our anonymous code release.

2.2 Multi-task learning balancers

A second body of work studies how to combine task losses or gradients once the architecture is fixed. Gradient-balancing methods rescale, project, or align per-task gradients; recent proposals along this axis include FAMO [7], which adapts loss weights from per-task progress, and Aligned-MTL [11], which aligns gradient directions through an independent-component decomposition. The dense-prediction survey [13] aggregates the line and reports that no balancer dominates across task pairs; the textbook tradeoff identified by Caruana [2] and quantified by Maninis et al. [9] through the Δ_m joint score remains the operating regime. Our setup uses a fixed static-weight aggregation with $\lambda_{\text{cat}} = 0.75$. We do not

propose a new balancer, and we ablate against FAMO and Aligned-MTL in Section 6 to check whether a drop-in balancer recovers the next-region cost we will report.

2.3 Prior POI multi-task work

A third line is concept-aligned with our task pair. The human-mobility representation model [3] jointly models attribute and trajectory signals through shared encoders, and STAN [8] addresses the next-location subproblem with spatio-temporal self-attention; we adapt STAN for our region head and add a training-fold-only additive $\alpha \log T$ prior in the spirit of GETNext [15], but our head is not a faithful reproduction of any one prior model. ReHDM [6] extends the hypergraph line with a region-aware dual-level model and serves as our concept-aligned external next-region baseline. Two recent works on the precise category and next-POI pairing are closer still. A first study trained a hard-parameter-sharing multi-task framework over POI-stable graph embeddings (DGI), comparing against single-task baselines such as HMRM [3] and MHA+PE [16], and reported only marginal joint gains, with most differences within standard deviations; the diagnosis pointed at representation mismatch, where a single shared encoder learns a compromise specialised enough for neither head [12]. A follow-up tested an embedding-side response by decomposing the monolithic graph embedding into independent spatial, temporal, and categorical sub-encoders, and recovered substantial category-side gains across three states, supporting the reading that the embedding choice was the primary factor [10]. Both responses kept the embedding's per-POI emission rate fixed and varied the *what* axis (which features to encode); our study varies the *granularity* axis instead.

Our work re-examines a default reading of multi-task learning for POI prediction: that joint training transfers signal between heads to lift both. With the substrate held fixed at the right granularity, joint cross-attention training delivers a small additional category lift at four of five Gowalla [4] state splits and pays a sign-consistent cost on next-region at every state. We use a controlled five-state ablation on user-disjoint folds to decompose the joint result, separating the substrate's contribution from the architecture's.

3 Method

We describe the substrate (the object of study), the backbone, and the training protocol. In this work, a check-in is defined as a tuple $c = (p, r, t)$ with POI p, region r, and timestamp t. Trajectories are split into non-overlapping length-nine windows predicting the category and region of the tenth check-in ($L = 9$).

3.1 The Check2HGI substrate

We propose **Check2HGI**, a check-in-level extension of HGI [5] that introduces a fourth emission level beneath POI: the hierarchy becomes Check-in $\rightarrow$ POI $\rightarrow$

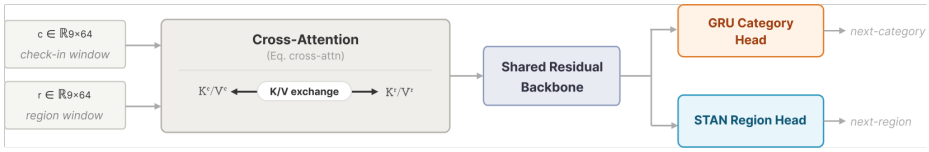

Fig. 1. Cross-attention multi-task backbone. Two task-specific input streams (X^c, X^r) feed a symmetric cross-attention block (Eq. 2); the bold black bidirectional K/V arrow is the cross-feed responsible for the encoder co-adaptation in Section 6.3, where a $\lambda = 0$ ablation does not isolate the targeted encoder. A shared residual backbone then splits into the GRU category head and the STAN-based region head with additive log-transition prior.

Region → City, so the same place visited twice can yield two distinct vectors. Visit nodes carry category and sin/cos timestamp features; edges follow each user's chronological sequence under exponential temporal decay. A two-layer GCN encodes check-ins; multi-head attention pools (*Checkin2POI*, then POI-to-region) plus a region-adjacency GCN propagate to region; an area-weighted sigmoid yields the city embedding.

Training maximises mutual information across the three adjacent boundaries via a per-boundary loss decomposition

$$\mathcal{L}_{\text{C2HGI}} = \alpha_{c2p}\,\mathcal{L}_{c2p} + \alpha_{p2r}\,\mathcal{L}_{p2r} + \alpha_{r2c}\,\mathcal{L}_{r2c}, \quad (\alpha_{c2p}, \alpha_{p2r}, \alpha_{r2c}) = (0.4, 0.3, 0.3), \tag{1}$$

where each $\mathcal{L}_b$ is a bilinear-discriminator binary cross-entropy against random-shuffle negatives at boundary b.

Substrates are pre-trained per state on the full state-level check-in graph; the user-disjoint protocol applies to the downstream classifier and the per-fold transition prior, not to the substrate's exposure (§4.1). Two products feed our backbone after the encoder is frozen, both 64-dimensional: a per-check-in vector $\mathbf{e}_c$ from the bottom level for the category stream, and a per-region vector $\mathbf{e}_r$ from the third level for the region stream.

3.2 Cross-attention multi-task backbone

The backbone keeps two parallel input streams bridged by cross-attention. Per-task encoders (two-layer MLPs with LayerNorm and dropout) project the input streams $X^c, X^r \in \mathbb{R}^{L\times d}$ into $H^c, H^r \in \mathbb{R}^{L\times d_m}$ ($L = 9$, $d = 64$, $d_m = 256$). With $\text{MHA}(Q, K, V)$ scaled dot-product attention ($h = 8$ heads),

$$\widetilde{H}^c = H^c + \text{MHA}(H^c,\, H^r,\, H^r), \qquad \widetilde{H}^r = H^r + \text{MHA}(H^r,\, H^c,\, H^c), \tag{2}$$

so each stream queries the other. The cross-attended streams pass through a four-block shared residual backbone (LayerNorm, linear, LeakyReLU, dropout) before splitting into per-task heads (Figure 1).

The category head is a two-layer GRU over the cross-attended check-in stream with attention pooling and a seven-class softmax. The region head is

an additive-prior STAN [8] variant: a STAN backbone over the cross-attended region stream, scored against learnable region embeddings, with an additive log-transition prior $\alpha \cdot \log T[r_{\text{last}}]$ (learnable α, built per fold from training-fold edges only). The log-prior pattern follows GETNext [15], but our head omits friendship and check-in graph priors and is not a faithful reproduction.

3.3 Training protocol

We aggregate the two task losses with a static weighting,

$$\mathcal{L} = \lambda_{\text{cat}}\, \mathcal{L}_{\text{cat}} + \lambda_{\text{reg}}\, \mathcal{L}_{\text{reg}}, \qquad \lambda_{\text{cat}} = 0.75, \quad \lambda_{\text{reg}} = 0.25, \tag{3}$$

where $\mathcal{L}_{\text{cat}}, \mathcal{L}_{\text{reg}}$ are unweighted cross-entropy, fixed across all MTL rows. The drop-in adaptive balancers in §6.2 (FAMO [7], Aligned-MTL [11]) do not improve over this static choice, supporting the simpler weighting; a λ_{cat} sensitivity sweep is in the anonymous code release.

We optimise the model with AdamW over three parameter groups (category, region, shared). The learning-rate schedule depends on state scale: OneCycleLR with per-task-group alternating updates at FL/CA/TX, and a per-head constant learning rate at AL/AZ. The alternating step rescues the region head from saturation at large transition graphs, but costs category-side signal at smaller scales. We use a batch size of 2048 (1024 at the largest state), gradient accumulation of 2, and 50 epochs in FP32; per-task best checkpoints are selected on validation Acc@10 and macro-F1 for epochs ≥ 5. Exact per-state settings are in the anonymous code release.

Folds, seeds, and multi-seed pooling follow the user-disjoint StratifiedGroupKFold setup of Section 4; the same splits are reused across substrates, heads, and recipes so paired tests are well defined. **Methodological caveat:** under cross-attention, $\lambda = 0$ ablation does not isolate the targeted encoder (it co-adapts through K/V); we use encoder-frozen isolation throughout (Section 6.3). Released artefacts (code, fold splits, seeds, per-fold transition priors, and regression tests) are listed in Section 1.

4 Experimental Setup

4.1 Datasets and preprocessing

We evaluate on five U.S.-state splits of the public *Gowalla* check-in dataset [4]: AL, AZ, FL, CA, TX. Each split is filtered to users with at least five check-ins, applied per state before fold creation so each state preserves its native cardinality. Per-state statistics are in Table 1; FL/CA/TX serve as the headline scale and AL/AZ as smaller-scale anchors. Region labels come from the per-state region partition used by the substrate engine.

For each user we form non-overlapping length-9 windows over the chronological sequence and predict the category and region of the tenth check-in (shorter sequences padded). Folds are **user-disjoint** via `StratifiedGroupKFold(groups=userid,`

Table 1. Dataset statistics for the five Gowalla [4] U.S.-state splits, after filtering to users with at least five check-ins. The headline three (FL/CA/TX) sit above the rule; the smaller-scale anchors (AL/AZ) sit below. Region cardinality is the prediction target on the next-region task.

State	Users	Check-ins	POIs	Regions	Mean traj. length
Florida (FL)	13,935	1,392,262	76,266	4,703	99.9
California (CA)	26,171	3,148,594	168,771	8,501	120.3
Texas (TX)	27,603	4,067,543	160,575	6,553	147.4
Alabama (AL)	1,622	109,695	11,666	1,109	67.6
Arizona (AZ)	3,331	227,956	20,469	1,547	68.4

`shuffle=True, random_state=42)` with five folds, so train/val/test never share a user. Splits are reused across substrates, heads, and recipes so paired tests are well defined. The next-region transition prior is rebuilt per fold from training-fold edges only and seed-tagged. The Check2HGI and HGI substrates are pre-trained on each state's full check-in graph under self-supervised mutual-information maximisation; the user-disjoint protocol applies to the downstream classifier and the per-fold transition prior, not to the substrate's exposure, so the setup is transductive at the substrate level.

4.2 Baselines and metrics

The headline contrast is the cross-attention multi-task model against matched-head single-task (STL) ceilings: a GRU category head and a STAN-based region head with an additive log-transition prior over training-fold-only edges [8]. **The same head classes appear on both sides of the contrast, so any difference between the multi-task and single-task rows is attributable to the sharing architecture, not to head capacity.** External baselines are POI-RGNN and MHA+PE (category) and Markov-1-region, single-task STAN, and ReHDM (region); we report them in §5.

Category metrics: macro-F1 (primary) and Acc@1/Acc@3 (top-k accuracy, secondary, since the 7-way distribution is skewed). Region metrics: Acc@10 (primary), Acc@5 and MRR (mean reciprocal rank, secondary). The joint score is Δ_m [9,13] against the matched-head STL ceiling, reported on a primary pair (macro-F1, MRR) and a secondary pair (macro-F1, Acc@10).

4.3 Statistical protocol

Paired contrasts use the paired Wilcoxon signed-rank test with `alternative='greater'` for directional claims and the two-one-sided test (**TOST**) for non-inferiority at $\delta \in \{2, 3\}$ pp. At the single-seed five-fold scale the max significance for $n = 5$ is $p = 0.0312$ (all folds agreeing).

Two ceiling protocols sit side by side. The **substrate axis** (Table 2) uses the seed-42 five-fold paired Wilcoxon at the $n = 5$ ceiling; multi-seed mean and seed-σ are shown for Check2HGI at AL/AZ/FL where multi-seed runs exist, but the paired test stays single-seed. The **architectural axis** (Tables 3, 4) pools across four extra seeds $\{0, 1, 7, 100\}$ on top of seed-42, yielding $n = 20$ multi-seed fold-pairs per state; the FL Δ_m pair additionally carries $n = 25$ fold-pairs from a second sweep. Effect sizes are in pp for F1, Acc@k, and MRR; Δ_m is in percent. Code, fold splits, seeds, and per-fold priors are in the anonymous snapshot referenced in §3.

5 Results

We report the empirical decomposition across five Gowalla state splits. Section 5.1 isolates the substrate axis (Check2HGI vs HGI under matched-head STL ceilings). Section 5.2 compares the cross-attention MTL backbone against those ceilings, the headline contrast. Section 5.3 reports the joint Δ_m score. Section 5.4 situates the model against external baselines.

5.1 Substrate ablation: per-visit context is task-asymmetric

Table 2(a) reports next-category macro-F1 under the matched-head GRU single-task ceiling, with substrate as the only changing factor. Check2HGI exceeds HGI by +14.18 to +29.02 percentage points (pp) at every state (smallest at AZ, largest at FL). The paired Wilcoxon signed-rank test reaches the $n = 5$ ceiling everywhere ($p = 0.0312$, 5/5 folds positive). Check2HGI macro-F1 is stable across four extra seeds $\{0, 1, 7, 100\}$ at AL/AZ/FL with seed-$\sigma \leq 0.17$ pp. Two robustness probes back this: the AL/AZ gap is head-invariant across linear, single-layer, GRU, and LSTM probes (probes in the anonymous code release), and the FL/CA/TX gap replicates under the matched-head ceiling. We read the lift as evidence that per-visit context, not POI-level attribute encoding, is what the next-category head exploits.

Table 2(b) reports Acc@10 under the matched-head STAN-based single-task ceiling. POI-stable HGI is nominally above Check2HGI by 1.6 to 3.1 pp at every state; on FL/CA/TX the gap shrinks to 1.6 to 2.1 pp and TOST non-inferiority at $\delta = 2$ pp passes at CA and TX. At FL the $\delta = 3$ pp test passes. AL and AZ stay outside both margins. The short-horizon transition structure driving the region head is captured equally well, or fractionally better, by per-place embeddings whose category-level pooling smooths the per-visit variance that helps the category head.

The substrate's effect is task-asymmetric. The category head reads per-visit context and lifts by an order of magnitude more than its standard-deviation envelope; the region head reads transition structure preserved under per-place pooling, tying within ± 2 pp. Section 6 attributes the asymmetry through a per-visit-context counterfactual replicated at Alabama and Arizona. Any multi-task system on this task pair operates on a substrate that already encodes the easier task more clearly than the harder one.

Table 2. Substrate ablation under matched-head single-task ceilings, five-fold cross-validation. Cells are seed-42 mean over folds ± fold-σ; Δ = Check2HGI cell − HGI cell at every row. The paired Wilcoxon at the $n = 5$ ceiling reaches $p = 0.0312$ when 5/5 folds agree. (a) Next-category macro-F1 with a GRU category head; the asterisk marks 5/5 folds positive. (b) Next-region Acc@10 with a STAN-based region head and additive log-transition prior built per fold from training-fold edges; negative Δ values indicate POI-stable HGI is nominally ahead, and TOST verdicts at $\delta \in \{2, 3\}$ pp report whether the gap stays inside the non-inferiority margin (✓) or not (×). The architectural axis (Tables 3 and 4) reports a multi-seed pooled ceiling for the architectural contrast: see §4.3.

(a) Next-category macro-F1.

State	HGI cat F1	Check2HGI cat F1	Δ (pp)	Wilcoxon p
Alabama (AL)	25.26 ± 1.18	40.76 ± 1.50	+15.50*	0.0312
Arizona (AZ)	28.99 ± 0.51	43.17 ± 0.28	+14.18*	0.0312
Florida (FL)	34.41 ± 1.05	63.43 ± 0.98	+29.02*	0.0312
California (CA)	31.13 ± 1.04	59.94 ± 0.59	+28.81*	0.0312
Texas (TX)	31.89 ± 0.55	60.24 ± 1.84	+28.34*	0.0312

(b) Next-region Acc@10.

State	HGI Acc@10	Check2HGI Acc@10	Δ (pp)	TOST $\delta = 2$	TOST $\delta = 3$
Alabama (AL)	61.86 ± 3.29	59.15 ± 3.48	−2.71	×	×
Arizona (AZ)	53.37 ± 2.55	50.24 ± 2.51	−3.13	×	×
Florida (FL)	71.34 ± 0.64	69.22 ± 0.52	−2.12	×	✓
California (CA)	57.77 ± 1.12	55.92 ± 1.20	−1.85	✓	✓
Texas (TX)	60.47 ± 1.26	58.89 ± 1.28	−1.59	✓	✓

5.2 Multi-task vs single-task on the per-visit substrate

Table 3 reports the headline contrast: the cross-attention multi-task backbone against the matched-head GRU and STAN-based single-task ceilings, both on Check2HGI, multi-seed pooled at $n = 20$ fold-pairs per state. Category is positive at four of five states: FL +1.40 pp ($p = 2\text{e}{-6}$), CA +1.68 pp ($p = 2\text{e}{-6}$), TX +1.89 pp ($p = 2\text{e}{-6}$), AZ +1.20 pp ($p < 1\text{e}{-4}$). At AL the contrast is statistically significant but small in magnitude ($\Delta = -0.78$ pp, $p = 0.036$, $n = 20$), under 2 % relative on the 41 % F1 scale. The four positive states sit outside the seed and fold envelope; we report the AL negative as is rather than absorbing it.

Next-region follows the cost-versus-gain pattern reported in the dense-prediction MTL line [13]. MTL pays −7.34 pp (FL) to −16.59 pp (TX) on Acc@10 at every state, all statistically significant ($p \leq 1.9\text{e}{-6}$). The pattern is broadly downward on AL → AZ → FL, with CA preserving the FL regime and TX breaking the monotonic shrink. We read the TX non-monotonicity descriptively: state-specific factors beyond raw class count (geographic POI distribution, user-density skew, per-state log-transition prior coverage) plausibly contribute; we do not read the AL → FL contraction as an inferential scaling claim.

Table 3. Headline contrast on the Check2HGI substrate: cross-attention multi-task model vs. matched-head single-task ceilings. Multi-seed pooled at $n = 20$ fold-pairs per state (seeds $\{0, 1, 7, 100\}$, five user-disjoint folds). Cells are mean $\pm$ pooled σ; Δ is multi-task minus single-task in percentage points; paired one-sided Wilcoxon p-values (alternative "greater" for positive Δ, "less" for negative). State codes: FL/CA/TX (headline scale, above the rule), AL/AZ (smaller-scale anchors, below).

	Next-category (macro-F1)				Next-region (Acc@10)			
State	STL	MTL	Δ	p	STL	MTL	Δ	p
FL	67.16±0.13	68.56±0.79	+1.40	2e−6	70.62±0.09	63.27±0.10	−7.34	1.9e−6
CA	62.39±0.13	64.07±0.14	+1.68	2e−6	56.85±0.09	47.35±0.11	−9.50	1.9e−6
TX	63.11±0.13	65.00±0.11	+1.89	2e−6	59.44±0.09	42.84±0.14	−16.59	1.9e−6
AL	41.35±0.17	40.57±0.24	−0.78	0.036	61.21±0.18	50.17±0.24	−11.04	1.9e−6
AZ	43.90±0.17	45.10±0.19	+1.20	<1e−4	53.06±0.15	40.78±0.07	−12.27	1.9e−6

Δ in percentage points (multi-task − single-task).

On Acc@10, the multi-task backbone trades the matched-head STAN-based single-task ceiling for a regime that sits below a one-step Markov-1-region floor at four of five states. Per-state Markov-1 / MTL Acc@10 (pp gap): AL 47.01/50.17 (+3.16); AZ 42.96/40.78 (−2.18); FL 65.05/63.27 (−1.78); CA 52.09/47.35 (−4.74); TX 54.94/42.84 (−12.10). AL is the only state above the floor. The matched-head Check2HGI ceiling sits +5 to +14 pp above Markov-1 at every state, so the cross-attention region head is trading that ceiling for the joint regime: under joint training, it underperforms on the short-horizon transition structure that a one-step Markov chain already captures. The Δ_m MRR axis at FL (+2.33 %, $n = 25$, statistically significant in Section 5.3) characterises the joint tradeoff more cleanly than Acc@10 in this regime; we report both axes and read them jointly.

With the substrate fixed at the per-visit granularity, the multi-task backbone delivers a small category lift at four of five states and pays a sign-consistent cost on the harder head at every state, the classic multi-task tradeoff. The cost is structural to the cross-attention shared-backbone interaction with the region head; Section 6 reports drop-in replacements (FAMO, Aligned-MTL, hierarchical-additive-softmax region head) that fail to close the next-region gap.

5.3 Joint Δ_m: broadly negative; FL on MRR the exception

We compute Δ_m [9,13] against the matched-head single-task ceiling. On the primary pair (category macro-F1, region MRR), FL carries an $n = 25$ multi-seed sweep and reaches Δ_m-MRR = +2.33 % at $p = 2.98\text{e}{-8}$, 25/25 fold-pairs positive: a Pareto-positive cell on the primary axis. The secondary pair at FL is Δ_m-Acc@10 = −1.12 % at $p = 3.20\text{e}{-5}$ (4/25); MRR rewards near-correct candidates that raw top-10 accuracy misses. That FL is exceptional on Δ_m-MRR is consistent with the Acc@10-saturated regime described above being what MRR sees through. The other four states report at the $n = 5$ Wilcoxon ceiling: Δ_m is negative on both axes (MRR from −24.84 % to −1.61 %; Acc@10 from −22.41 %

Table 4. External-baseline comparison at FL/CA/TX. Italicised rows are matched-head STL ceilings on Check2HGI; the proposed row is our MTL model. POI-RGNN and MHA+PE are user-disjoint reproductions (published POI-RGNN [1] used non-user-disjoint folds, so absolute values are not directly comparable). STAN is the STAN backbone [8] on our HGI substrate with our training-fold-only log-prior. STAN/ReHDM are at FL only above the headline scale (ReHDM scales quadratically in region cardinality); a literature-faithful STAN reproduction is omitted on page-budget grounds (§7).

	Category macro-F1			Region Acc@10		
Method	FL	CA	TX	FL	CA	TX
External baselines (faithful reproductions)						
Markov-1 (floor)	23.98±1.22	19.97±1.77	18.17±1.38	65.05±0.93	52.09±0.80	54.94±0.46
POI-RGNN [1]	34.49±1.14	31.78±0.82	33.03±0.70	–	–	–
MHA+PE [16]	32.06±0.23	29.13±0.71	29.91±0.43	–	–	–
STAN [8] (HGI)	–	–	–	**73.58±0.43**	**60.45±0.97**	**62.70±0.37**
ReHDM [6]	–	–	–	65.68±0.26	–†	–†
Matched-head single-task ceilings (Check2HGI substrate)						
STL category ceiling	67.16±0.13	62.39±0.13	63.11±0.13	–	–	–
STL region ceiling	–	–	–	70.62±0.09	56.85±0.09	59.44±0.09
Proposed cross-attention multi-task model						
MTL (ours)	**68.56±0.79**	**64.07±0.14**	**65.00±0.11**	63.27±0.10	47.35±0.11	42.84±0.14

Bold cells: best per column. The row label **MTL (ours)** marks the proposed model. † ReHDM not reported at CA/TX: the dual-level hypergraph scales quadratically in region cardinality, exceeding our compute budget.

to −6.85 %), at most 1/5 fold-pairs positive. FL is the only Pareto-positive cell on the primary axis at multi-seed.

5.4 External baselines

Table 4 situates the model against external baselines. On next-category, our matched-head single-task ceiling on Check2HGI sits well above our user-disjoint POI-RGNN reproduction [1], but the gap should be read with care: the published POI-RGNN evaluation used non-user-disjoint folds, so absolute numbers are not directly comparable. The substrate-axis ablation of Section 5.1 attributes a substantial share of the lift to per-visit context, not to multi-task aggregation; the matched-head STL category ceiling on Check2HGI (67.16 at FL) roughly doubles the strongest user-disjoint POI-RGNN reproduction (34.49), and §6.1 attributes ∼72% / ∼64% of the AL/AZ substrate gap specifically to per-visit context. On next-region, the matched-head STAN-based head with additive log-prior on the HGI substrate is the strongest external row at this scale; per-place HGI is a more natural fit for the region head, consistent with the substrate-axis reading. Our multi-task row trails this row by −10 to −20 pp on Acc@10, consistent with the substrate gap compounded by the cross-attention region cost.

6 Mechanism and Ablations

This section turns the empirical decomposition into a mechanism reading, stress-tests the next-region cost against drop-in alternatives, and closes with a methodological note on cross-attention loss-side ablations.

6.1 Per-visit context counterfactual (five states)

To isolate per-visit context, we compare canonical Check2HGI against POI-pooled Check2HGI (canonical vectors mean-pooled per POI and applied uniformly to every visit, killing per-visit variation while preserving the training signal). Canonical vs. pooled isolates per-visit context; pooled vs. HGI isolates the contrastive training signal.

Under the matched-head GRU single-task ceiling, the substrate gap (canonical − HGI) splits consistently at all five states (Figure 2): per-visit context dominates at every state, with both components positive at every state. Paired Wilcoxon signed-rank tests reach the $n = 5$ ceiling at AL and AZ ($p = 0.0312$, 5/5 folds positive each); the FL/CA/TX pattern is consistent with the same monotone ordering at all five per-fold pairs. A two-band pattern emerges across state scale: at small states (AL ∼72 %, +11.19 pp of +15.50; AZ ∼64 %, +9.08 of +14.18) the training-signal residual is 28–36 %; at large states (FL ∼90 %, +26.06 pp of +29.02; CA ∼89 %, +26.08 of +29.41; TX ∼90 %, +25.42 of +28.16) the pooled embeddings collapse almost entirely to the HGI level, leaving ∼10 % to the training signal. The pattern is consistent with large transition graphs (4.7k–8.5k regions) providing weaker per-POI contrastive signal, so Check2HGI's advantage at large states is almost entirely in the per-visit contextual variation it introduces. A head-free linear probe at AL gives a comparable per-visit share (∼63 %; the GRU head amplifies the share rather than creating it).

6.2 Drop-in MTL ablations do not recover the next-region cost

The next-region cost is sign-consistent across all five states (7 to 17 pp on Acc@10). At FL we tried three drop-in replacements on the same folds and seeds with substrate, backbone, optimiser and schedule fixed at §3: a magnitude balancer (FAMO [7]), a direction-aligning balancer (Aligned-MTL [11]) and a hierarchical-additive-softmax region head. None reached a +3 pp Acc@10 recovery target or paired-Wilcoxon $p_{\text{greater}} < 0.10$ (FAMO +0.62, Aligned-MTL −0.11 on reg and −0.90 on cat, hierarchical head ≈ +1.7). The cost stays intact, suggesting it is structural to the cross-attention interaction with the region head rather than a loss-recipe artefact; five-state sign-consistency is supporting context, not direct multi-state evidence.

6.3 Cross-attention $\lambda = 0$ is not encoder isolation

A natural ablation zeros one task's weight in Eq. (3), on the reading that $\lambda_{\text{cat}} = 0$ or $\lambda_{\text{reg}} = 0$ ablates the corresponding encoder. Under cross-attention this is unsound: the targeted encoder still updates through the key/value projections of

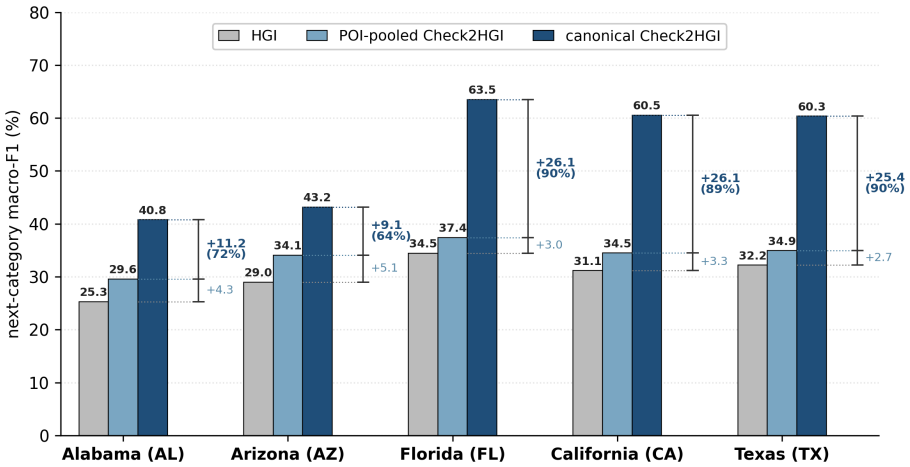

Fig. 2. Per-visit context decomposition at five states under the matched-head GRU STL ceiling. The substrate gap (canonical Check2HGI − HGI) splits into a per-visit component (canonical − POI-pooled) and a training-signal component (POI-pooled − HGI). Per-state shares are reported in §6.1.

the surviving stream, so its parameters co-adapt to the surviving task. A targeted probe at AZ on the per-head constant LR recipe quantifies this: encoder-frozen isolation reaches region Acc@10 60.72, while the loss-side $\lambda_{\text{cat}} = 0$ ablation reaches 62.70 on the same folds and seeds, a +1.98 pp gap the surviving stream drives through the targeted encoder's K/V projections. We therefore use **encoder-frozen isolation** (targeted encoder random-init, frozen) as the clean architectural decomposition; the note generalises to any cross-task attention multi-task model that ablates by zeroing a task weight, and regression tests are in our anonymous code release.

7 Discussion and Limitations

The headline reading is that the multi-task picture is dominated by the *substrate's task-asymmetric value*, not cross-task transfer through joint training. Per-visit context is statistically significant for next-category at every state (+14 to +29 pp matched-head); on next-region the same substrate ties or marginally trails per-place HGI by 1.6 to 3.1 pp under the matched-head single-task ceiling, with TOST $\delta = 2$ pp passing at California and Texas. With the substrate fixed, the cross-attention multi-task backbone adds a small category lift at four of five states (Arizona, Florida, California, Texas) and is statistically significant but small in magnitude at Alabama ($\Delta = -0.78$ pp, $p = 0.036$, $n = 20$, under 2% relative); on next-region it pays a sign-consistent 7 to 17 pp cost on Acc@10 at every state. The conventional reading where joint training transfers

signal to lift both heads does not hold on the harder task: the category lift survives the substrate ablation, and the region cost survives the substrate fix. The substrate's task-asymmetric value is large in absolute terms (the matched-head STL category ceiling on Check2HGI, 67.16 at FL, roughly doubles the strongest user-disjoint POI-RGNN reproduction at 34.49), and is recoverable by any downstream architecture that consumes per-visit context. The honest reading is that the substrate comes first (task-asymmetric) and the multi-task coupling second (tradeoff-paying).

Four limitations apply. The per-visit decomposition replicates at AL+AZ but is not generalised to the remaining three states, and the drop-in panel ran at Florida only. On Acc@10 the multi-task backbone trails the Markov-1-region floor at four of five states (§5.2); we read this as a structural cost of joint training on the harder head rather than a Florida-specific saturation, and report Δ_m MRR alongside (FL +2.33 %, $n = 25$, highly significant). External coverage is asymmetric: ReHDM [6] is reported at FL only above the headline scale and a literature-faithful STAN reproduction is omitted due to page constraints (Table 4 caption). The substrate is transductive at the state level (§4); held-out-user substrate retraining, cross-task transfer (e.g. next-POI), and architectural extensions for the region head are left to follow-up work.

8 Conclusion

We introduced Check2HGI, a check-in-level contextual graph embedding, and tested whether it carries the multi-task POI prediction win across five U.S.-state Gowalla splits. The substrate is task-asymmetric: per-visit context lifts matched-head next-category macro-F1 by +14 to +29 pp at every state, with the AL/AZ counterfactual attributing ∼72 % and ∼64 % of the gap to per-visit context; per-place embeddings tie or marginally exceed Check2HGI on next-region. With the substrate fixed, joint multi-task training adds a small category lift at four of five states and pays a sign-consistent 7 to 17 pp region cost; drop-in balancers and a head-side fix do not close the gap. Three open lines follow: asymmetric routing across the shared backbone to relax cross-attention interference, encoder enrichment fusing per-visit contrast with the explicit spatial-temporal sub-encoders of [10], and a region-side dynamic prior conditioned on the user's recent trajectory.

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